

EXACT DISTRIBUTION OF THE QUOTIENT OF CERTAIN INDEPENDENT GENERALIZED HYPERGEOMETRIC VARIABLES

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1. Introduction

Let X be a random variable whose frequency function is,

$$(1.1) \quad f(x; b; a, \beta, \gamma, \mu) = \begin{cases} C x^{\gamma-1} {}_1F_1(a; \beta; -bx^\mu) & x > 0, b > 0, a > 0, \beta > 0, \gamma > 0, (a-\gamma/\mu) > 0, \\ & (\beta-\gamma/\mu) > 0, \mu > 0 \\ 0 & \text{otherwise.} \end{cases}$$

where ${}_1F_1(a; \beta; -bx^\mu)$ is the confluent hypergeometric function we consider, and

$$C = \mu b^{(\gamma/\mu)} \Gamma(a) \Gamma(\beta - \gamma/\mu) / \Gamma(\beta) \Gamma(a - \gamma/\mu) \Gamma(\gamma/\mu).$$

Form (1.1) is a generalization of the generalized Gamma distribution. The familiar Gamma, generalized Gamma, Chi, Chi-squared, exponential and Weibull variates are special cases, as are certain functions of normal variates. The authors of the R.A.D.C. Reliability Notebook (1961) recognize that the heart of the reliability problem is that no distribution can be safely assumed. Therefore it is fairly commonplace in reliability analysis to encounter data which is incompatible with generalized Gamma, exponential, Weibull and other familiar probability models. Such data motivates research to enlarge the group of probability distribution which are useful in reliability analysis. Dwivedi (1969) has shown that the generalized hypergeometric distribution is a closer fit to the data of Menon (1965) than generalized Gamma distribution. Here in this article we derive the exact distribution of the quotient of independent generalized hypergeometric variables in terms of the H-function [4]. It is shown that the result of Malik (1967) is a special case of it.

2. The distribution of the ratio of two independent random variables

Let $X_i, i=1,2$ be random variables independently distributed with the following density functions,

$$(2.1) \quad f(x_i; b_i; a_i, \beta_i, \gamma_i, \mu_i) = \begin{cases} C_i x_i^{\gamma_i-1} {}_1F_1(a_i; \beta_i; -b_i x_i^{\mu_i}) & \text{for } x_i > 0 \\ 0 & \text{otherwise.} \end{cases}$$

where $i=1,2$ and

$$(2.2) \quad C_i = \mu_i b_i^{\gamma_i/\mu_i} \Gamma(a_i) \Gamma(\beta_i - \gamma_i/\mu_i) / \Gamma(\beta_i) \Gamma(a_i - \gamma_i/\mu_i) \Gamma(\gamma_i/\mu_i).$$

Let $U = X_1/X_2$ where X_1 and X_2 be two independent random variables. The density function of U , denoted by $g(u)$, is obtained by using the properties of the Mellin transform [3]. Since the mathematical expectation of a product of two independent random variables is the product of their expectations, when the expectations exist, the Mellin

transform of a product is the product of the Mellin transforms, when the variables are statistically independent. After obtaining the Mellin transform, the density function of U denoted as $g(u)$ is obtained by obtaining the inverse Mellin transform.

Now consider the $(s-1)^{\text{th}}$ moment of U ; we have,

$$(2.3) \quad E(U^{s-1}) = C_1 \int_0^\infty x_1^{s-1} x_1^{\gamma_1-1} {}_1F_1(a_1; \beta_1; -b_1 x_1^{\mu_1}) dx_1 \times \\ C_2 \int_0^\infty x_2^{s-1} x_2^{\gamma_2-1} {}_1F_1(a_2; \beta_2; -b_2 x_2^{\mu_2}) dx_2$$

$$(2.4) \quad = C_1 C_2 \Gamma(\beta_1) \Gamma[(\gamma_1 + s - 1)/\mu_1] \Gamma[a_1 - (\gamma_1 + s - 1)/\mu_1] \times \\ \Gamma(\beta_2) \Gamma[\gamma_2 - s + 1]/\mu_2 \Gamma[a_2 - (\gamma_2 - s + 1)/\mu_2] / \\ \mu_1 \Gamma(a_1) \Gamma[\beta_1 - (\gamma_1 + s - 1)/\mu_1] b_1^{\gamma_1/\mu_1} b_1^{(s-1)/\mu_1} \mu_2 \Gamma(a_2) \times \\ \Gamma[\beta_2 - (\gamma_2 - s + 1)/\mu_2] b_2^{\gamma_2/\mu_2} b_2^{(s-1)/\mu_2}.$$

where C_1 and C_2 are given by (2.2) for $i=1,2$ respectively.

Substituting the values of C_1 and C_2 in (2.4) we have,

$$(2.5) \quad E(U^{s-1}) = \int_0^\infty u^{s-1} g(u) du \\ = A_1 \Gamma[(\gamma_1 + s - 1)/\mu_1] \Gamma[a_1 - (\gamma_1 + s - 1)/\mu_1] \times \\ A_2 \Gamma[\gamma_2 - s + 1]/\mu_2 \Gamma[a_2 - (\gamma_2 - s + 1)/\mu_2] / \\ \Gamma[\beta_1 - (\gamma_1 + s - 1)/\mu_1] b_1^{(s-1)/\mu_1} b_1^{(1-s)/\mu_1} \mu_2 \Gamma[\beta_2 - (\gamma_2 - s + 1)/\mu_2]$$

where

$$(2.6) \quad A_1 = \Gamma[\beta_1 - (\gamma_1/\mu_1)] / \Gamma(\gamma_1/\mu_1) \Gamma[a_1 - (\gamma_1/\mu_1)]$$

and

$$(2.7) \quad A_2 = \Gamma[\beta_2 - (\gamma_2/\mu_2)] / \Gamma(\gamma_2/\mu_2) \Gamma[a_2 - (\gamma_2/\mu_2)]$$

Denoting by $h(s)$ the product of Gammas containing s on the right-hand side of (2.5) we have,

$$(2.8) \quad E(U^{s-1}) = A_1 A_2 (b_1^{1/\mu_1} / b_2^{1/\mu_2}) h(s) (b_2^{1/\mu_2} / b_1^{1/\mu_1})^s.$$

Now by considering the inverse Mellin transform we obtain the density function $g(u)$ of the random variable U as follows,

$$(2.9) \quad g(u) = A_1 A_2 (b_1^{1/\mu_1} / b_2^{1/\mu_2}) \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} h(s) [(b_1^{1/\mu_1} / b_2^{1/\mu_2})^s u]^{-s} ds$$

$$(2.10) \quad = A_1 A_2 (b_1^{1/\mu_1} / b_2^{1/\mu_2}) \times H_{3,3}^{2,2} \left[(b_1^{1/\mu} / b_2^{1/\mu_2}) u \right. \\ \left. \begin{array}{l} (1 - \alpha_1 + [\gamma_1 - 1] / \mu_1, 1 / \mu_1), (1 - [\gamma_2 + 1] / \mu_2, 1 / \mu_2), (\beta_2 - [\gamma_2 + 1] / \mu_2, 1 / \mu_2) \\ ([\gamma_1 - 1] / \mu_1, 1 / \mu_1), (\alpha_2 - [\gamma_2 + 1] / \mu_2, 1 / \mu_2), (1 - \beta_1 + [\gamma_1 - 1] / \mu_1, 1 / \mu_1) \end{array} \right]$$

Thus we have proved the following theorem.

Theorem 3.1

Let X_1 and X_2 be two independent random variables with density functions as given in (2.1) for $i=1,2$ respectively. Then $U = X_1/X_2$ has the density

$$(2.11) \quad g(u) = \{ \Gamma(\beta_1 - \gamma_1 / \mu_1) \Gamma(\beta_2 - \gamma_2 / \mu_2) / \Gamma(\gamma_1 / \mu_1) \Gamma(\gamma_2 / \mu_2) \times \\ \Gamma[a_1 - (\gamma_1 / \mu_1)] \Gamma[a_2 - (\gamma_2 / \mu_2)] \} (b_1^{1/\mu_1} / b_2^{1/\mu_2}) \times \\ H_{3,3}^{2,2} \left[(b_1^{1/\mu_1} / b_2^{1/\mu_2}) u \right. \\ \left. \begin{array}{l} (1 - a_1 + [\gamma_1 - 1] / \mu_1, 1 / \mu_1), (1 - [\gamma_2 + 1] / \mu_2, 1 / \mu_2), (\beta_2 - [\gamma_2 + 1] / \mu_2, 1 / \mu_2) \\ ([\gamma_1 - 1] / \mu_1, 1 / \mu_1), (\alpha_2 - [\gamma_2 + 1] / \mu_2, 1 / \mu_2), (1 - \beta_1 + [\gamma_1 - 1] / \mu_1, 1 / \mu_1) \end{array} \right]$$

Corollary 3.1:

Let $\alpha_1 = \beta_1$ and $\alpha_2 = \beta_2$ in (2.11). Then (2.11) reduces to the following,

$$(2.12) \quad g_1(u) = [\Gamma(\gamma_1 / \mu_1) \Gamma(\gamma_2 / \mu_2)]^{-1} (b_1^{1/\mu_1} / b_2^{1/\mu_2}) \\ H_{1,1}^{1,1} \left[(b_1^{1/\mu_1} / b_2^{1/\mu_2}) u \mid \begin{array}{l} (1 - [\gamma_2 + 1] / \mu_2, 1 / \mu_2) \\ ([\gamma_1 - 1] / \mu_1, 1 / \mu_1) \end{array} \right].$$

Now expanding the H-function using the result of Braaksma (1964), we get

$$(2.13) \quad H_{1,1}^{1,1} \left[(b_1^{1/\mu_1} / b_2^{1/\mu_2}) u \mid \begin{array}{l} (1 - [\gamma_2 + 1] / \mu_2, 1 / \mu_2) \\ ([\gamma_1 - 1] / \mu_1, 1 / \mu_1) \end{array} \right] \\ = \sum_{\nu=0}^{\infty} \{ \Gamma[(\gamma_2 + 1) / \mu_2 + (\mu_1 / \mu_2)(\gamma_1 - 1) / \mu_1 + \nu] / \\ \nu! (1 / \mu_1) \} (-1)^\nu \left[(b_1^{1/\mu_1} / b_2^{1/\mu_2}) u \right]^{\mu_1 [(\gamma_1 - 1) / \mu_1 + \nu]}$$

Now when $\mu_1 = \mu_2 = \mu$ we have

$$(2.14) \quad H_{1,1}^{1,1} \left[(b_1 / b_2)^{1/\mu} u \mid \begin{array}{l} (1 - [\gamma_2 + 1] / \mu, 1 / \mu) \\ ([\gamma_1 - 1] / \mu, 1 / \mu) \end{array} \right] \\ = \sum_{\nu=0}^{\infty} \{ \Gamma[(\gamma_2 + 1) / \mu + ((\gamma_1 - 1) / \mu) + \nu] / \nu! (1 / \mu) \} (-1)^\nu [(b_1 / b_2)^{1/\mu} u]^{(\gamma_1 - 1) + \mu \nu}$$

$$(2.15) \quad = \sum_{\nu=0}^{\infty} \{ \Gamma[\gamma_2 + \gamma_1]/\mu + \nu \} / \nu! (1/\mu) (-1)^\nu [(b_1/b_2)^{1/\mu} u]^{(\gamma_1-1) + \mu\nu}$$

$$(2.16) \quad = \mu \Gamma[(\gamma_2 + \gamma_1)/\mu] [(b_1/b_2)^{1/\mu} u]^{\gamma_1-1} [1 + (b_1/b_2) u^\mu]^{-(\gamma_1 + \gamma_2)/\mu}.$$

Therefore for $\alpha_1 = \beta_1$, $\alpha_2 = \beta_2$ and $\mu_1 = \mu_2 = \mu$ (2.12) can be written as follows

$$(2.17) \quad g_1(u) = \mu \Gamma(\gamma_1/\mu + \gamma_2/\mu) (b_1/b_2)^{1/\mu} [(b_1/b_2)^{1/\mu} u]^{\gamma_1-1} / \\ \Gamma(\gamma_1/\mu) \Gamma(\gamma_2/\mu) [1 + (b_1/b_2) u^\mu]^{(\gamma_1 + \gamma_2)/\mu}.$$

Now by taking out $(b_1/b_2) u^\mu$ from $[1 + (b_1/b_2) u^\mu]^{-(\gamma_1 + \gamma_2)/\mu}$ we have,

$$(2.18) \quad g_1(u) = \mu \Gamma(\gamma_1/\mu + \gamma_2/\mu) (b_1/b_2)^{\gamma_2/\mu} [u]^{-\gamma_2-1} / \\ \Gamma(\gamma_1/\mu) \Gamma(\gamma_2/\mu) [1 + b_2/b_1 u^{-\mu}]^{(\gamma_1 + \gamma_2)/\mu}, \mu > 0.$$

It is seen that the density function $g_1(u)$ in (2.18) is the same as obtained by Malik (1967).

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